UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) Sixth Semester - Spring 2023

Roll No. ....

Paper: Computational Physics II | Course Code: PHYS-3504 | Time: 3 Hrs. | Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1 Write short answer of the following questions: (6x5=30)

I. Explain with example the following terms in MTLAB
   a. polyval()
   b. roots()
   c. factor()

II. Write syntax with example for the followings in MATLAB:
   a. Extrapolation of data
   b. Curve fitting

III. Write a MATLAB program with function to calculate $f(x)$, given by:

$$f(x) = 28x^3 - 15x^2 + 7x - 3$$

Calculate, print and plot values for x and $f(x)$. Also find minimum and maximum of $f(x)$ values for fifteen different values of x.

IV. Write MATLAB to plot 'x' against f(x) values using subplot such that: $x = [0\ 4\pi]$, $y(x) = n*\sin(x)$ and n=1,2,3,4

V. Write MATLAB program to determine corresponding force values for the work and distance values as give in the table. Also calculate total force and average work values.

w (work)	20	27	47.5	62
d(distance)	12	27	41	56

VI. Two vectors are given:

u= 5i+9j-2k and v= 4i-6j+12k

Use MATLAB to calculate the dot product u.v of the vectors in two ways:

a) Define u as row vector and v as a column vector, and then use matrix multiplication.

b) Use the dot function.

Solve the following.

Q.2. (a) Write a MATLAB program to plot x vs ψ such that:

$$\psi_n(x) = A\frac{n\pi x}{L}$$

Where $A = \sqrt{\frac{2}{L}}$, L = 4, n = [1:4] **(6 Marks)**

(b) Write MATLAB program to evaluate the $\int \left(\frac{\sqrt{x}}{6}\right)dx$ by Trapezoidal Rule. (n=6) **(4 Marks)**

Q.3. Write MATLAB program to describe motion of a charged particles in magnetic field. Also draw estimate output graphs with proper curve labels, x & y labels and title. **(10 Marks)**

Q.4. Create matrix D in MATLAB such that

$$D = \begin{bmatrix} 10 & 9 & 8 & 7 & 6 \\ 12 & 14 & 16 & 18 & 20 \\ 1 & 4 & 7 & 10 & 13 \\ 25 & 8 & 11 & 2 & 1 \\ 33 & 14 & 29 & 6 & 12 \end{bmatrix}$$

(i) Write v1 equals 4^{th} column of D.
(ii) find location of value 18 in v1, (iii) create a linear vector k containing all elements of D. (iv) create v2 equals 3^{rd} row of D. (v) determine u= v1.v2 (vi) plot 4^{th} column against 1^{st} row of D (vii) select and sort in ascending order the 4^{th} and 5^{th} rows of D. (viii) select a 3x3 middle matrix as a set of D (10

Computational-Physics (II)

ADP spring 2023

Q#01 write short answer of following questions:

~~a(I):~~

a- Polyval():-

⇒ **Discription:-** returns the value of polynomial at point 'x'.

⇒ **syntax:-**

polyval(p, x)

⇒ **Example:-** % let's say you have a polynomial $3x^2+2x+1$ & evaluate this polynomial at $x=2$.

```
>> p=[3 2 1];
>> x= 2;
>> y= polyval(p,x)
>>disp('evaluate polynomial:');
>>disp('y');
```

b- roots():-

⇒ **Discription:-** roots(C) computes the roots of polynomial whose co-efficients are the elements of the vector C.

⇒ **syntax:-**

roots (C).

⇒ **Example:-**

% polynomial is $2x^3-6x^2+2x-1=0$ and find its roots:-

```
>> p=[2 -6 2 1];
>> r= roots(p);
>> disp('roots of polynomial:');
>> disp('r');
```

b- factor():-

⇒ **Discription:-** F=factor(x), returns all irreducible factors of x in vector F. If x is an integer, factor returns the prime factorization of x. If x is a symbolic expression, factor returns subexpressions that are factors of x.

⇒ **syntax:-**

F=factor(x)

⇒ **Example:-**

% find the prime factor of number 60.

```
>> n=60;
>> f=factor(n);
>> disp('Factor of n:');
>> disp(f);
```

(II)

a- Extrapolation of data:-

⇒ **Descriptions:-**

Extrapolation is the process of estimating unknown values by extending a known set of data.

here, x & y are known dat points
xi = points at which to entrapolate.
method:- linear, nearest, spline, pchip.
'extrap':- specifies that entrapolation should be used for points outside range of x.

⇒ **syntax:-**

```
yi = interp1( x,y, xi, method, 'extrap')
```

⇒ **Examples:-** %. Take x & y from user and extrapolate the value at x=6.

```
% known data process:-
>> x=[1 2 3 4 5];
>> y=[2 4 6 8 10];
>> xi = 6;
>> yi = interp1(x,y, xi, 'linear', 'extrap');
>> disp(yi);
```

b- Curve fitting:-

⇒ **Description:-** curve fitting is the process of constructing a curve or mathematical function that has the best fit to a series of data points.

⇒ **syntax:-** fitresult = fit(x, y, fitType)

x: The independent variable data (vector).

y: The dependent variable data (vector)

fit Type: A string specifying type of model to fit (e.g. 'poly1' for a linear fit, 'poly2' for a quadratic fit.

⇒ **Example:-** % Taking x & y from user that appears to have a linear relationship, and fit a straight line to it.

% example data:

```
>> x = [1  2  3  4 5];
>> y = [2.2  4.1  5.9  8.2  10.1];
>> fitresult = fit( x , y , 'poly1');
>> disp( fit result);
```

% plot data & the fit:

```
>> plot (fitresult, x, y);
>> x label('x');
>> ylabel('y');
>> title ( 'linear fit');
>> legend ( 'Data', 'Fitted curve');
```

(III)

% Define given data

```
>> x = [-10 : 15 : 10];
>> f(x) = 28*x.^3 -15*x.^2+7*x-3;
```

```
>>y(x)=polyval(f(x),x);
% plot data
>> plot(y(x));
>> xlabel('values of x');
>> ylabel('value of polynomial');
>> title('plot of y(x)');
>> grid on;

% minimum of f(x) for x:
>> [min-y, min-index]= min(y(x));
>> disp('min of y(x):');
>> disp(min-y);
>> disp(min-index);

% maximum of f(x) for x.
>> [max-y, max-index] = max(y(x));
>> disp('max of y(x):');
>> disp(max-y);
>> disp(max-index);
end
```

(V)

% define given data

```
>> W = [20, 27, 47.5, 62];
>> d = [12, 27, 41, 56];
```

% Find force

```
>> F = W./d;
```

% Find Total-Force

```
>> Total_Force = sum(F);
```

% Find avg-work

```
>> avg-work = mean(W);
```

% display results

```
>> disp('Force:');
>> disp(F);
>> disp('Sum of Force:');
>> disp(Total-Force);
```

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```
>>disp('average of work');
>>disp( avg-work);
```

(VI)

% define given data

```
>> u=[5 , 9 , -2];
>> v=[4; -6; 12];
```

% Dot product by matrix multiplication

```
>> Dot-product = u*v;
>>disp('Dot product of u & v');
>>disp( Dot-product);
```

% Dot product by dot function

```
>> Dot-p = dot(u,v);
>> disp('Dot product by dot function:');
>> disp( Dot-p);
```

Solve the following

Q#2:-

(a):

```
% define given data
>> y=(A*n*pi*x.)/L;
>> x = linespace(-10,10,4);
>> L= 4;
>> A = sqrt (2/L);

% plot data
>>plot (x, y, '--o');
>> xlabel( 'x-axis');
>> ylabel('y-axis');
>> title( 'plot x vs y');
>> grid on;
end
```

(b):

% Tapezoidal method

```
>> Y = sqrt(x)/6;
>> a = 1 ;
>> b = 6;
>> Trapezoidal-Y=Trapz(Y,x, a, b);
>> disp( 'Trapezoidal w.r.t x');
>> disp( Trapezoidal-Y);
end
```

Q#3:- % given data.

```
>> q =1 ;  >> m=1;  >> h=0.5;  >>tmax=100;
>>  x=100;  >>y=20;  >> Vo=5;  B=0.5;
>> w = (q*B)/m;
>> ind= 1 ;
>> t=1;
while ( t<=tmax)
>> ta (ind)=t;
>> xa (ind)=x;
>> ya (ind)=y;
>> Vx = Vo* sin (w*t);
>> Vy = Vo * cos(w*t);
>> ax = w* Vy;
>> ay = -w* Vx;
>> Vx = Vx+ax*h;      >>Vy = Vy+ay*h;
>> x=x+Vx*h;          >> y=y+ Vy*h;
>> t=t+h;             >> ind=ind+1;
end
```

% display result

```
>>disp('time x y');
>> tb(:,1)=ta';
>>tb(:,2)=xa';
>> tb(:,3)=ya';
>>disp(tb);
```

% plot & label ta, xa.

```
>> plot(ta,xa, '--o');
>> hold on;
>> xlabel('time');
>> ylabel('position:x');
>> title('plot of ta vs xa');
>> grid on;
```

% plot & label ta, ya.

```
>> plot(ta, ya, '-o');
>> x label('time');
>> y label ('position-y');
>> Title ('plot of ta vs ya');

>> grid on;
>> legend ('Time ,position-x, position-y');
end
```

Q#4:-

% create matrix D as given

```
>> D = [10 9 8 7 6; 12 14 16 18 20; 1 4 7 10
        13; 25 , 8, 11, 2 , 1; 33, 14, 29, 6, 12];
```

(a) % create V1 equal to 4th column of D

```
>> V1 = D(:, 4);
```

(b) % location of value 18 in vin

```
>> index = find (V1 = = 18);
```

(c) % create a linear vector K containing all elements of D

```
>> K = D(:);
```

(d) % create V2 equal to 3rd row of D

```
>> V2 = D(3, :);
```

(e) % Determine u

```
>> u = dot(V1, V2);
```

(f) % plot 4th column against 1st row of D.

```
>>plot( D(:, 4), D(1, :));
>> xlabel( '4th column of D');
>> ylable( '1st row of D');
>> title ('plot of 4th column against 1st row');
>> grid on;
```

(g) % select & sort in ascending order

```
>> sorted-rows = sort( D([4, 5], :), 2);
```

(h) % 3 X 3 middle matrix

```
>> middle-matrix = D( 2:4, 2:4);
```

Q(IV)

% define given data

```
>> x = [0 : pi/10 : 4*pi]
>> for n = 1:4;
>> y = sin(n*x);
>> subplot(1, 4, n);
>> plot ( x, y);
>> xlabel('x');
>> ylabel('y');
>>title('plot x vs y');
end
```